Formula 1 Podium Prediction

Milestone 1 — Analytical Report

Team 11 — Rawan Shahin (58-0516) · Omar Elghazouly (58-2262) · Nouran Ahmed (58-0948) · Rawan Osama (58-1750)

Data: Kaggle / Ergast Formula 1 World Championship dataset (1950–2024, 14 tables) · Notebook: "F1 Podium Prediction – Milestone 1 – Team 11" (Kaggle, version 13, runs top to bottom in about 15–20 minutes on CPU)

In one paragraph. For every driver in every race we predict whether he finishes in the top 3, using only what is known after qualifying and before the start. We cleaned the 14 linked tables (sentinels, types, keys, one row per driver and race), answered the three data-engineering questions, built 20 leakage-safe features in 6 groups for **every season 1950–2024**, and split the data by season (train 1950–2018, validation 2019–2021, test 2022–2024). Of 13 modelling attempts, a tuned **XGBoost** model was best on validation (PR-AUC 0.745) and reached **PR-AUC 0.711 on the unseen 2022–2024 seasons**, against 0.647 for the simple rule "podium if grid ≤ 3". Training on the whole history costs nothing measurable compared with recent seasons only, and helps the final model (Section 7). The starting grid carries most of the signal; recent form, standings and qualifying add to it. All explanations describe what the model relies on, **not** what causes podiums.

1. Task, target and prediction cutoff

- **Unit of analysis:** one driver entering one race; the pair (raceId, driverId) is unique in every modelling row.
- **Target:** `podium = positionOrder ≤ 3` (binary). `positionOrder` is never missing, so every row has a label.
- **Cutoff:** after qualifying, before the race starts. Results, points, status, laps, pit stops, lap times and standings *after* the race are never features. Standings tables for race X already include race X, so only the previous round is used.
- **Class balance:** 13.6% of starters finish on the podium (14.4% from 2003). A model that always says "no" would be about 86% accurate and useless — this is why **PR-AUC is the main metric** and ROC-AUC the secondary one (Decision 3); F1, precision and recall are reported at a threshold chosen on validation.

2. Data audit and cleaning

Loading (Decision 13). All 14 CSVs are read as-is, so the sentinel `\N` stays visible as text in `raw` (160,076 cells in 6 tables) together with 68 empty cells in `qualifying.q2/q3`. Cleaning turns all 160,144 markers into real missing values (NaN) on copies; `raw` is never changed (checked by a fingerprint at the end of cleaning and again after Section 9).

Semantic typing (Decision 14). 24 columns are converted in place: 10 to numbers, 7 to dates, and 7 lap or pit times to seconds with a helper `time_to_seconds()`, which equals `milliseconds / 1000` on all 589,081 laps and 11,371 pit stops.

Keys. 41 checks (14 primary keys, 4 natural keys, 23 foreign keys): 0 empty keys, 0 duplicates, 0 broken links.

Rows removed (applied in this order):

Step	Rule and reason	results rows
start	raw `results`	26,759
Decision 6	Drop the 11 Indianapolis 500 races (1950–1960): a separate series — only 0.6% of its drivers raced another GP that season, 99.8% American (would inflate Q2)	26,354
Decision 8	Drop drivers who never started (`positionText` F = failed to qualify, W = withdrew): 1,704 rows. Keep the 87 pit-lane starters (grid = 0). Removing W rows is tested in Section 7 (no change to the model).	24,650
Decision 5	Shared drives (1950s): drop every row of a repeated (raceId, driverId) pair. The grid cannot tell the cars apart, and "keep the first row" would choose by finishing order (resultId follows it in 1,074 of 1,125 races) — leakage	**24,525**

Cleaning deliberately does **not** fill missing values or drop rows with gaps: filling belongs in the model Pipeline, fitted on training data only, and `dropna()` would remove every driver who did not finish.

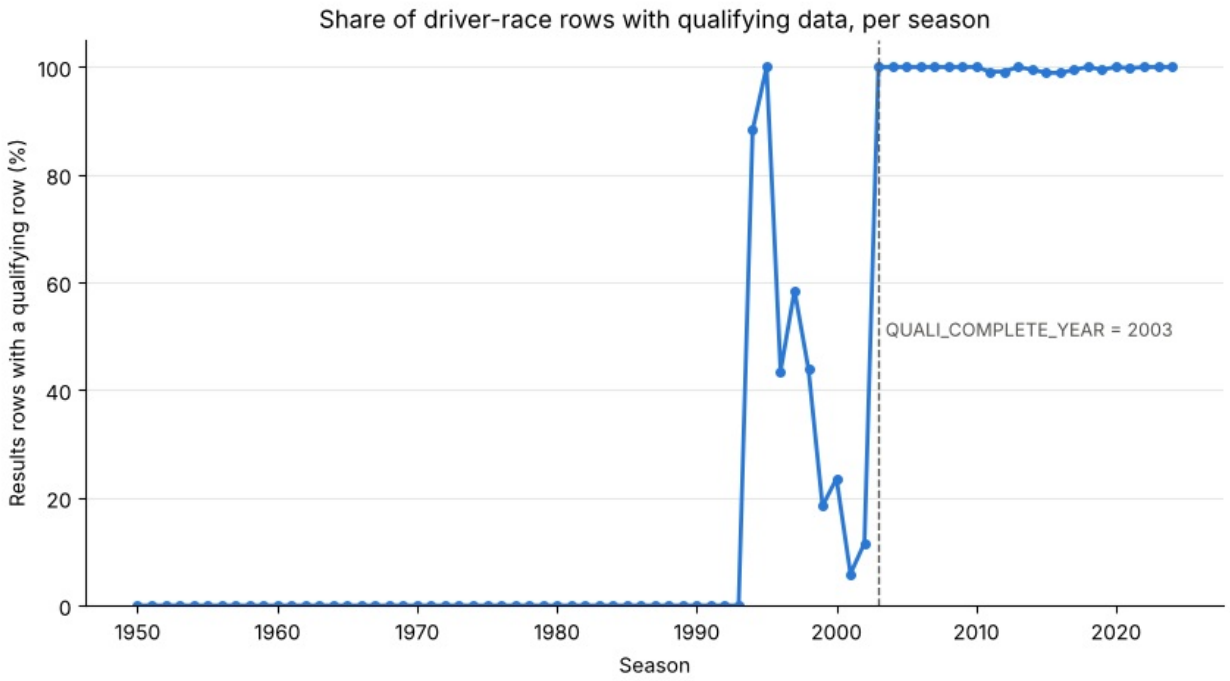


Figure 1. Qualifying coverage per season: none before 1994, patchy 1996–2002 (down to 6% in 2001), at least 98.9% from 2003. The model still trains on all seasons (Decision 4): in older rows the qualifying features are missing and are filled inside the Pipeline with a “was missing” flag (Decision 22).

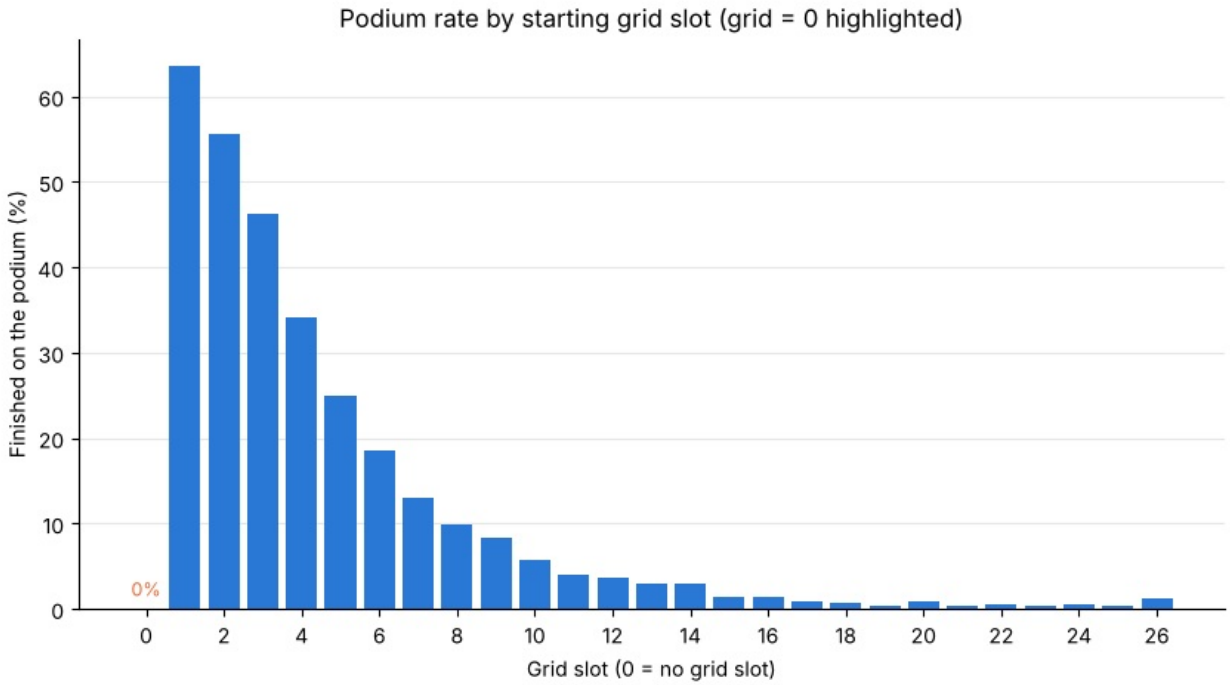


Figure 2. Podium rate by grid slot (raw data): 63.5% from pole, under 6% from 10th, 0% for grid = 0 — the strongest single signal and the evidence for the grid features and the “grid ≤ 3” rule baseline.

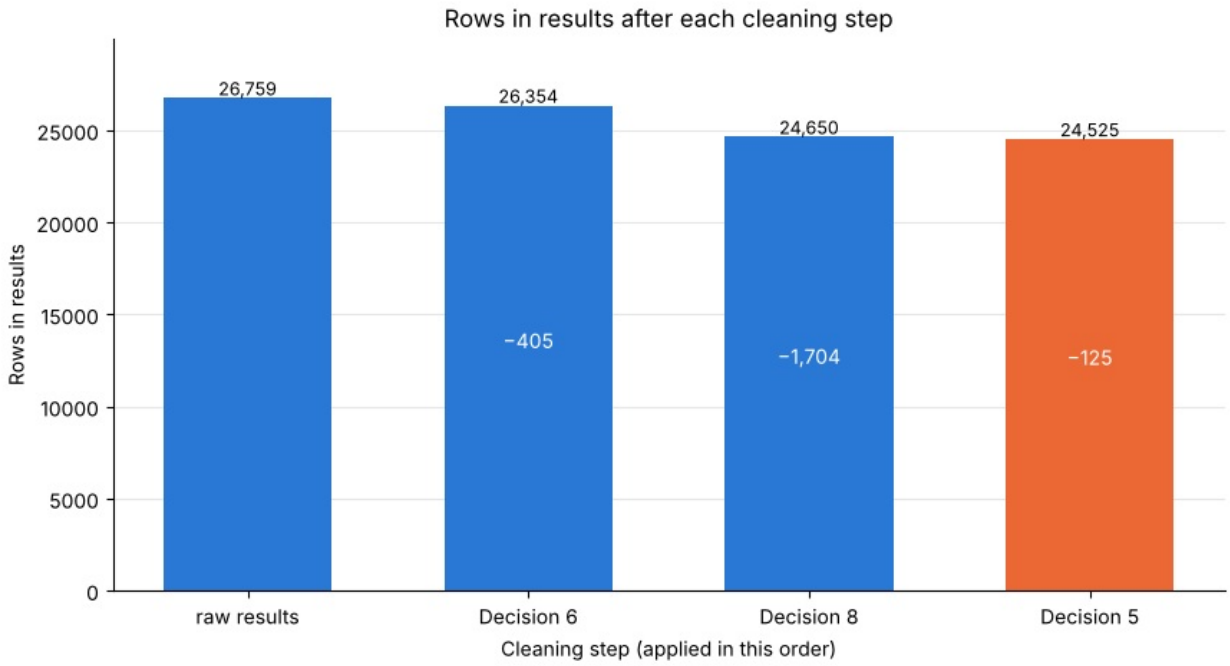


Figure 3. EDA after cleaning: rows per table before and after cleaning. Five of the 14 tables lose rows (only Decisions 6, 8 and 5 remove rows).

EDA before vs after. The grid = 0 spike shrinks from 1,638 rows (6.1%) to the 87 pit-lane starters; entries exceeded starters by up to 12.7 drivers per race (1989) and after cleaning every row is a starter; the podium share of rows rises from 12.7% to 13.6%. Retirement rates fell from 40–50% of starters before 2000 to about 12% in the 2020s — eras differ strongly, which is why we split by time and test in Section 7 whether the older eras help or hurt the model.

3. Data-engineering questions

All three use one joined table (`analysis`: results → races → circuits, drivers, constructors, status; 24,525 rows, grain asserted, every join many-to-one).

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Definition: front row = grid slot 1 or 2 (per the TA: the question is about the starting position, not qualifying); rate = podiums ÷ front-row starts, per circuit, before 2014 vs from 2014. **Grid vs qualifying:** they differ in 23.3% of driver-races since 1994 (1.5% in 1994–2003, 25.4% in 2004–2013, 30.4% from 2014, because penalties move drivers back); 3.8% of drivers who qualified on the front row did not start there. **Sample rule (Decision 15):** only circuits with at least 10 front-row starts in each period are ranked (16 circuits), with 95% Wilson intervals.

Answer. Over all circuits, front-row conversion rose from **56.7%** before 2014 to **76.2%** from 2014. Highest from 2014: Monaco 89.5%, Suzuka 88.9%, Abu Dhabi 86.4%; lowest: Budapest 59.1%, Mexico City 61.1%, Sakhir 66.7%. The ranking **changes a lot** (Spearman −0.08, bootstrap interval −0.50 to 0.38; 0.36 with 2003–2013 as the "before" period) — Monaco climbs from 15th to 1st, Budapest falls from 2nd to 16th — but with only 10–26 starts per circuit after 2014, only Monaco's change is clearly larger than chance. *Grain:* driver-race. *Missingness:* grid is complete. *Cutoff:* the grid is known before the start.

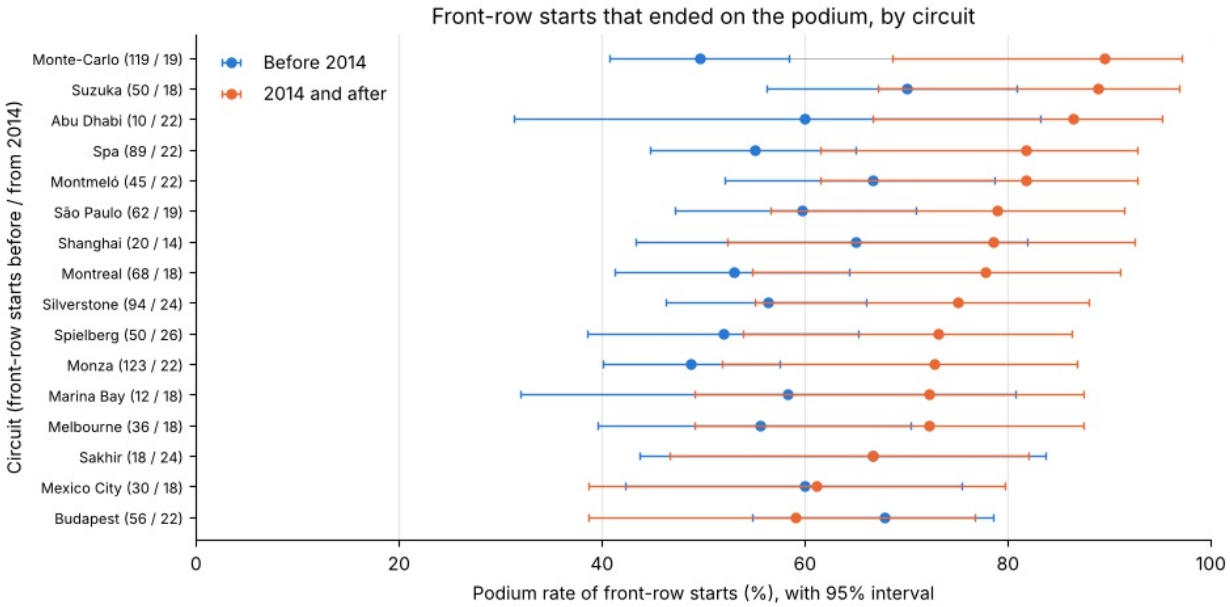


Figure 4. Q1: front-row → podium rate per circuit, before 2014 (blue) vs from 2014 (orange), with 95% Wilson intervals.

Q2. Do drivers podium more often at home, after controlling for starting position?

Home flag (Decision 12): an explicit nationality → country dictionary (43 nationalities, all mapped and asserted; dual nationalities are home in either country; one raw value "Argentinian " had a trailing space; Las Vegas is spelled "United States"). Home race in 1,914 of 24,525 rows (7.8%). **Design (Decision 16):** compare home and away drivers *inside* four grid buckets (1–3, 4–6, 7–10, 11+), so drivers who started from similar places are compared; pit-lane starts are left out; one pooled difference re-weights away drivers to the home drivers' grid mix; the interval comes from a bootstrap over whole races.

Answer. Inside the buckets the rates are nearly identical (grid 1–3: 57.1% home vs 55.8% away; 4–6: 26.9% vs 26.1%; 7–10: 7.2% vs 9.3%; 11+: 1.1% vs 1.7%). The adjusted difference is **−0.5 points** (95% interval −1.7 to +0.8); for 2003–2024 −1.4 (−4.1 to +1.5). There is **no sign of a home advantage** once the starting position is controlled (an association, not a causal claim). *Cutoff:* nationality, circuit and grid are all known before the race.

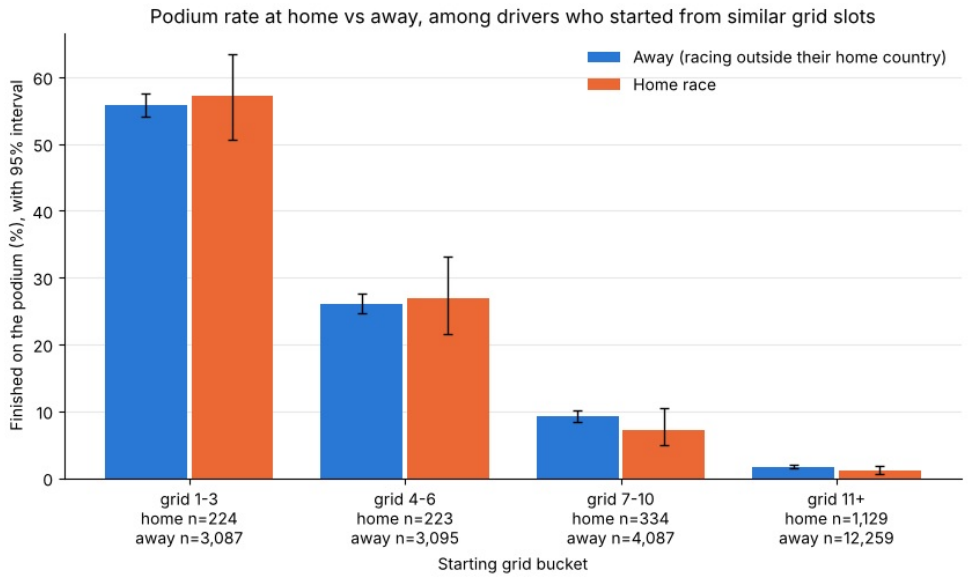


Figure 5. Q2: podium rate of home (orange) vs away (blue) drivers inside each grid bucket, 95% Wilson intervals; sample sizes in the axis labels.

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Status mapping (Decision 11): 65 statuses among 2014–2024 starters, all mapped. Fixed by the TA: Puncture, Tyre, Tyre puncture, Fire, Safety belt = mechanical; Refuelling = not mechanical. Our rule for the rest: failure of a named car component (Engine, Gearbox, Power Unit, Brakes, Hydraulics …) = mechanical; collisions, accidents, spins, disqualifications = not; 9 unclear statuses (Retired, Wheel, Damage, Debris, Out of fuel …, 77 starts) are not mechanical in the main answer and counted in a “broad” check. Renamed teams (Decision 7) are linked into lineages (e.g. Force India → Racing Point → Aston Martin) for Q3 only. Rate = mechanical retirements ÷ starts.

Answer. Overall the rate fell from 9.2% to 5.5% of starts. 2014–2021: highest **RB / Toro Rosso 14.7%** (47 of 319), Alpine / Renault 13.5%, McLaren 13.0%; lowest Mercedes 4.1%. 2022–2024: highest **Alpine 10.4%** (14 of 135), Sauber / Alfa Romeo 8.8%, RB 5.9%; lowest Aston Martin, McLaren and Red Bull (3.7% each). 8 of 10 teams improved; intervals separate the periods only for McLaren and (barely) RB. The answer is robust: strict vs broad mapping rank the teams almost identically (Spearman 0.91 / 0.95, same top 3). *Grain:* starts. *Cutoff:* status is post-race and is never a model feature.

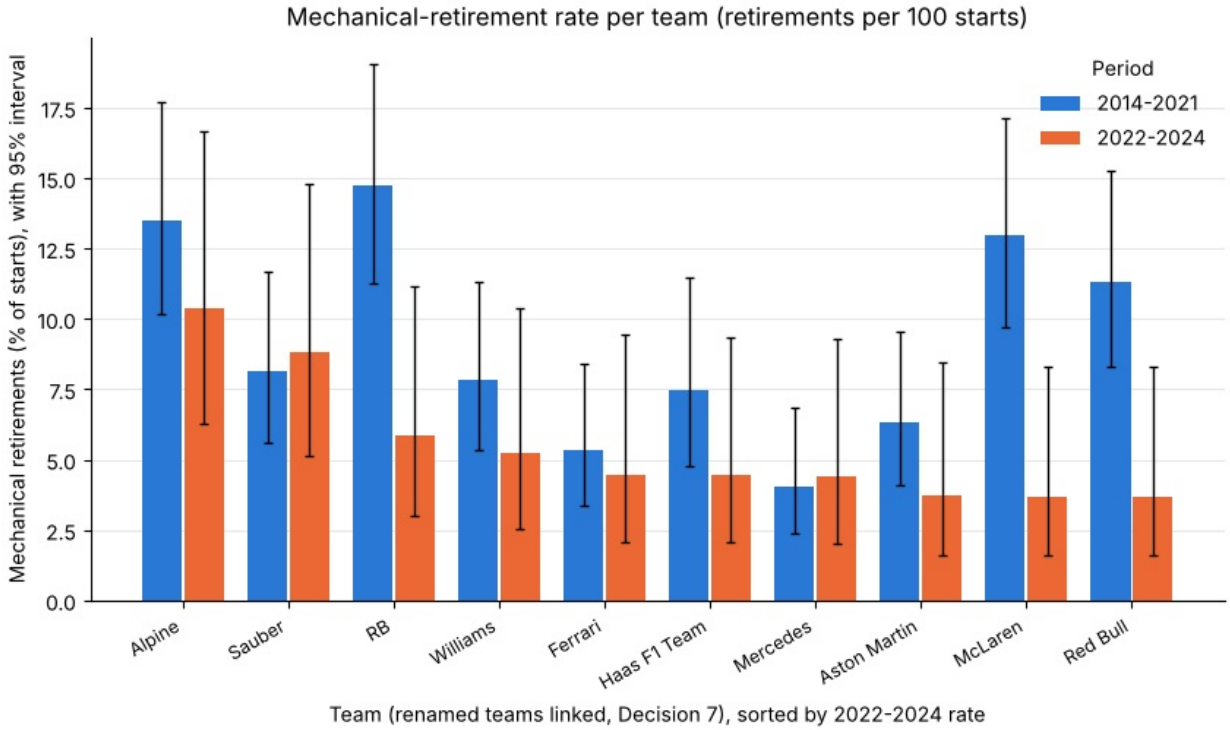


Figure 6. Q3: mechanical-retirement rate per team (lineage), 2014–2021 vs 2022–2024, 95% Wilson intervals.

4. Features and why each is valid at the pre-race cutoff

Modelling table (Decision 4): `model_df` = **24,525 rows (1,114 races, 1950–2024)**, podium rate 13.6%, one row per (raceId, driverId) (asserted). The TAs asked us to use the entire dataset, so the model learns from every era; whether this helps or hurts is tested in Section 7. Every feature only looks at earlier races. Missing values stay NaN and are filled inside the model Pipeline (training medians + "was missing" indicators, Decision 22) — the main gap is qualifying, missing in 57.5% of all rows (almost all before 2003). Feature-justification charts and trials use only seasons up to 2021 (train + validation) — the test seasons are never looked at.

Group	Feature	What it is	Why it is known before the start
grid	`grid_position`	starting slot; pit lane = last slot + 1 (Decision 17)	the grid is fixed before the start
	`grid_front_row`, `grid_pitlane_start`	flags for slot 1–2 and pit-lane start	from the grid
quali	`quali_position`	qualifying position	qualifying ends before the race
	`quali_q1_gap_pct`	% slower than the fastest Q1 time of the race	qualifying ends before the race
	`quali_grid_penalty`	grid slot − qualifying position (> 0 = moved back)	both known before the start
drv	`drv_podium_rate / avg_finish / dnf_rate _last10`	driver's results in his previous 10 races	`.shift(1)`: earlier races only
con	`con_podium_rate / avg_finish / dnf_rate _last10`	team's results (both cars) in its previous 10 races	`.shift(1)`: earlier races only
stand	`stand_driver / con _points_before, _rank_before`	championship points and position before the race	standings of the previous round
ctx	`ctx_home_race`	circuit in the driver's home country (Decision 12)	nationality and circuit are fixed
	`ctx_field_size`, `ctx_driver_age`, `ctx_driver_races_before`	starters, age on race day, earlier starts	known when the grid forms / fixed

Feature decisions.

- **Decision 17 — pit-lane starters** (87 rows, 0 podiums): `grid_position` = last slot + 1 plus a flag. Raw 0 would sit in front of pole; NaN would be filled with a median (about 10th) — both wrong. (A TA confirmed this encoding on Piazza.)
- **Decision 18 — form window:** last 10 races, across season boundaries, restarting after a break of more than a year (otherwise Mercedes' first 2010 race would get its 1954–55 results as "form"). Trial on 1950–2021, ROC-AUC of the podium rate alone (driver / team): N = 3: 0.778 / 0.811, 5: 0.813 / 0.830, **10: 0.835 / 0.842**, 15: 0.841 / 0.844, 20: 0.841 / 0.842 — 10 is within 0.01 of the best and reacts within a season. No history → NaN (not 0, which would claim "never on the podium").
- **Decision 19 — constructor form follows `constructorId`:** the Q3 lineage map covers only 2014+ renames; linking would treat renames two ways, and both versions differ in only 0.3% of rows with the same AUC (0.842).
- **Decision 20 — no year / era feature:** with a time split every test season is unseen; the podium base rate per season (11.4–18.9%, 1950–2021) follows the field size, which is already a feature. Modelling the eras explicitly is left for the next milestone (TA answer).
- **Decision 9 — sprint races: use partly.** 18 sprint weekends (2021–2024, 1.4% of rows). The sprint is finished before the Sunday start, so the Sunday grid (which the sprint set in 2021–22) is known at prediction time — not leakage. We keep the races and the grid but build no sprint features (they would be NaN in 98.6% of rows, all in validation/test). Standings from the previous round never include the current weekend's sprint points.
- **Qualifying outlier:** one Q1 time of 1,002.6 s (1995) is not a real lap; a guard (Q1 above 1.5 × the fastest) sets such gaps to NaN (asserted).

Leakage checks (8.12): driver and team form recomputed by brute force from races with an earlier date equal the features on 300 random rows; standings: "points before + race points = standings after" holds in 99.7% of normal-weekend rows (proving the table includes the race itself; the exceptions are 1950–1990 seasons where only a driver's best results counted); asserts that no post-race column is a feature.

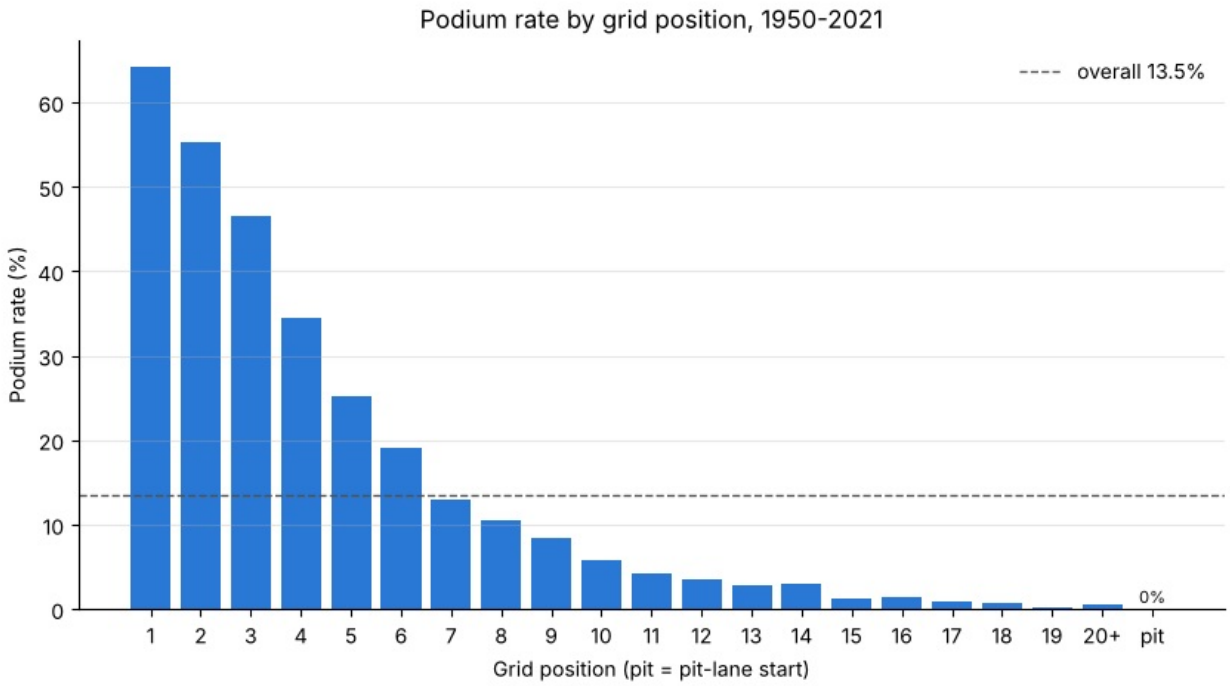


Figure 7. Podium rate by grid position, 1950–2021: 64.1% from pole, under 6% from 10th; pit-lane starts 0%.

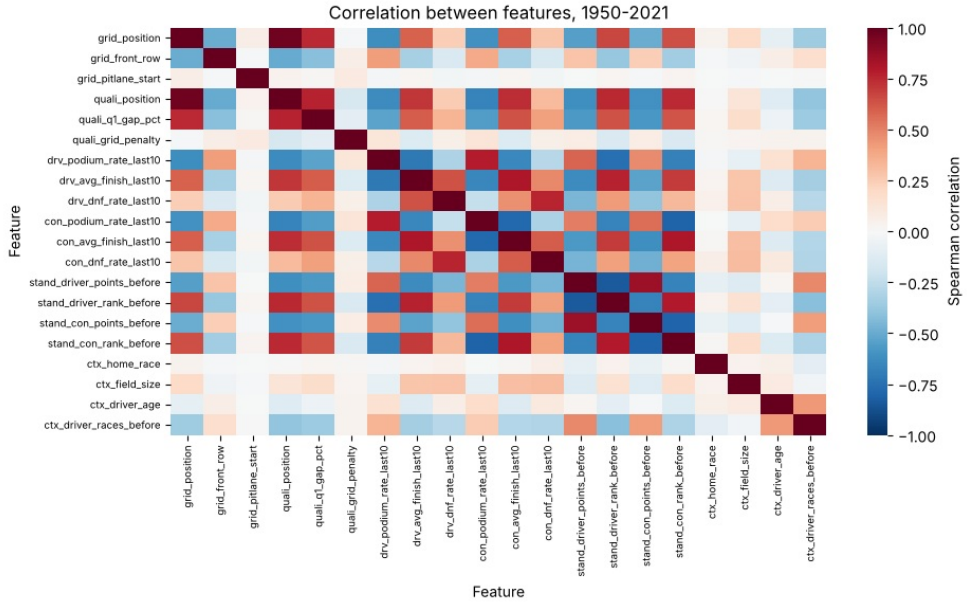


Figure 8. Spearman correlation between the 20 features (1950–2021): strong overlap within and between groups (grid vs qualifying position 0.97).

The other group charts (notebook 8.13b–f) show the same monotone patterns: within 0.25% of the fastest Q1 time → 55.5% podium; a driver with no podium in his last 10 races → 2.9%, with 7 or more → about 60%; team podium share above 0.5 per car → 56.0%; championship leader before the race → 60.0%; experience peaks at 151–200 earlier starts (27.5%).

5. Train / validation / test split (Decision 10)

We predict future races from past ones, so the split is by season. A random 70/15/15 row split would put drivers of 100% of the test races into training too, and 99.9% of training rows would come from races later than the first test race. Three boundary options were compared (the brief's "≤ 2019 / 2020–21 / ≥ 2022" is only a naive example, per the TA):

Option	Train	Validation	Test	Train rows	Val rows	Test rows
A	1950–2019	2020–2021	2022–2024	22,398	775	1,352
B (chosen)	**1950–2018**	**2019–2021**	**2022–2024**	**21,978**	**1,195**	**1,352**
C	1950–2019	2020–2022	2023–2024	22,398	1,214	913

Podium share is 15.1% in every validation and test split and 13.4–13.5% in training (older races had bigger fields), so balance does not decide. Option B gives a validation set of three seasons in which the short, unusual COVID season 2020 (17 races at 14 circuits) is 28% of the rows instead of 44% (option A), and keeps the whole 2022 ground-effect rules era as an

untouched test set — an honest “forecast after a rule change”. Final sizes: train 21,978 rows (986 races, 13.4% podium), validation 1,195 (60 races, 15.1%), test 1,352 (68 races, 15.1%).

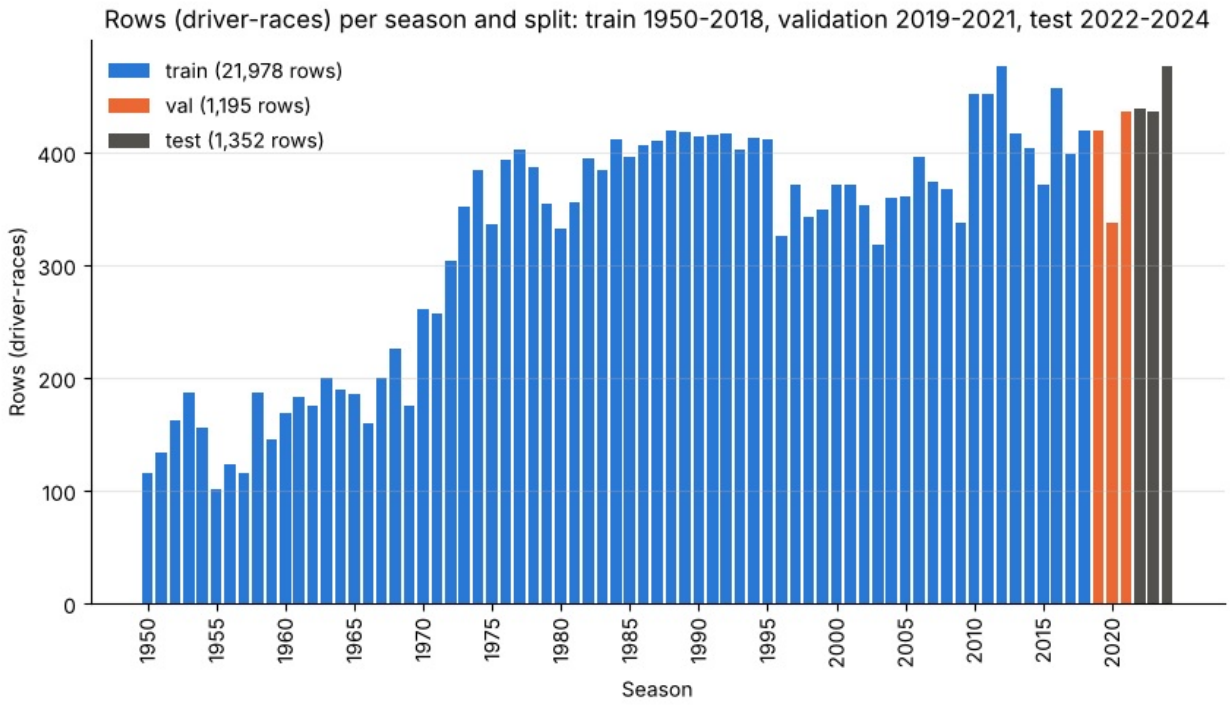


Figure 9. Rows per season and split: train 1950–2018 (blue), validation 2019–2021 (orange), test 2022–2024 (grey).

6. Models and results

Every model is a scikit-learn `Pipeline`: median imputer with “was missing” indicators (fitted on train only), plus a scaler for logistic regression and the network (Figure 10: 51,649 missing training cells in 15,025 rows → 0, 20 → 31 columns, spreads differing about 2,000× → all standard deviation 1). XGBoost handles missing values itself. For every model the probability threshold is the one with the best F1 on validation, applied unchanged to test.

- **Rule to beat:** “podium if grid ≤ 3” (score = 1 / grid position).
- **Logistic regression** (baseline, per the TA): weighted sum of scaled features → sigmoid probability; regularised by C.
- **Random forest:** 300 decision trees on random samples of rows and features, probabilities averaged.
- **XGBoost:** gradient boosting — small trees built one after another, each correcting the remaining errors.
- **Shallow neural network (Keras):** 1–2 hidden ReLU layers + sigmoid output, Adam optimiser, dropout and early stopping on the last 15% of training rows (validation seasons stay unused).

Each family has three labelled attempts (default → tuned on validation → class weights / dropout), 13 attempts in total.

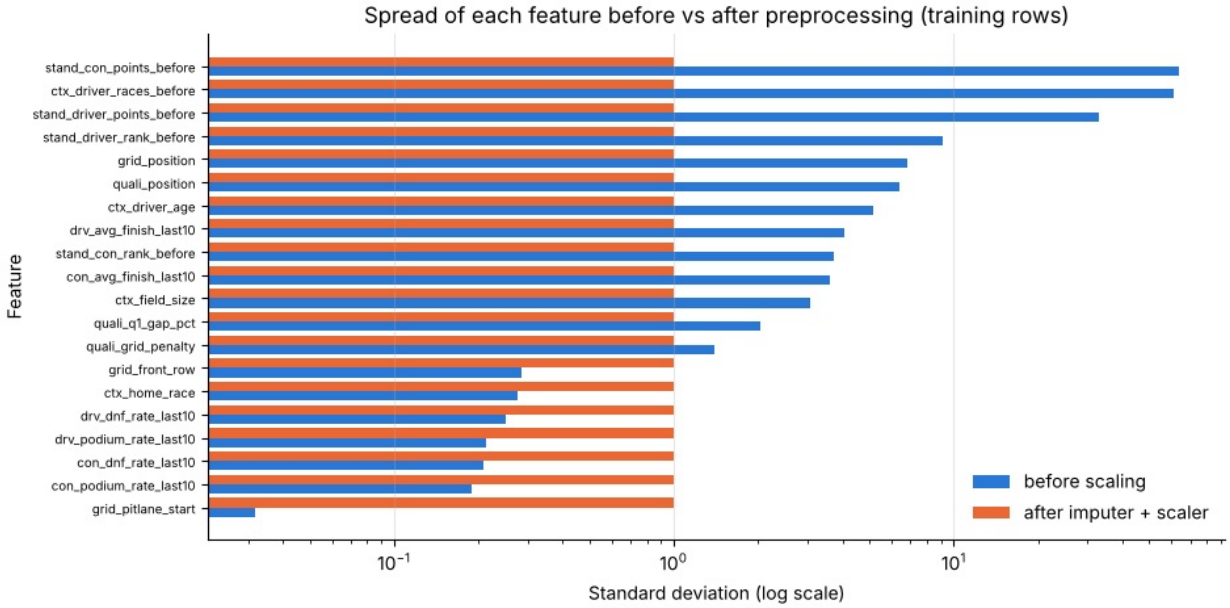


Figure 10. Effect of the preprocessing on the training rows: standard deviation of each feature before (blue) and after imputation and scaling (orange), log scale.

Model (best attempt of each family)	Train PR-AUC	Val PR-AUC	Val ROC-AUC	Val F1	Test PR-AUC	Test ROC-AUC	Test F1	Test precision	Test recall
Rule: grid ≤ 3	0.497	0.683	0.917	0.719	0.647	0.913	0.649	0.650	0.647
Logistic regression (LR_baseline)	0.573	0.729	0.937	0.732	0.718	0.936	0.630	0.667	0.598
Random forest (RF_v2_tuned)	0.762	0.743	0.938	0.741	0.716	0.939	0.679	0.638	0.725
XGBoost (XGB_v2_tuned) — final	**0.593**	**0.745**	**0.939**	**0.735**	**0.711**	**0.940**	**0.639**	**0.691**	**0.593**
Neural network (NN_v3_tuned)	0.588	0.717	0.933	0.743	0.717	0.935	0.681	0.658	0.706

All 13 attempts with all five metrics on train, validation and test are in notebook cell 10.11. Notable attempts: the default forest overfits (train PR-AUC 1.000, validation 0.699) until leaves hold at least 10 rows (0.743); the default XGBoost (depth 4) reaches 0.737 and the tuned one (depth 2, learning rate 0.03) 0.745; class weights never improve PR-AUC (LR 0.726, RF 0.731, XGB 0.744); dropout helps the network (0.709 → 0.713 → 0.717 tuned). Train PR-AUC is *lower* than validation for most models because the training seasons include the older eras, which are harder to predict from pre-race data — not a sign of overfitting.

Decision 21 — final model and threshold: `XGB_v2_tuned` (300 trees, depth 2, learning rate 0.03, minimum child weight 1), the attempt with the highest validation PR-AUC; threshold **0.476** (best validation F1: precision 0.731, recall 0.739). The top three attempts are within 0.002 of each other, so the pre-agreed rule decides. “Top 3 probabilities per race” was compared on validation (F1 0.717) but not used, because the inference function scores one driver at a time.

Test result. On the 2022–2024 seasons the final model reaches **PR-AUC 0.711** (validation 0.745), ROC-AUC 0.940, F1 0.639: 121 of 204 podiums found with 54 false alarms. The drop from validation is expected — the test seasons are a new rules era — and every learned model still beats the grid rule by about 0.06–0.07 PR-AUC. The models are close on test (0.711–0.718), so the ranking between them is within noise; the final model was not re-chosen after seeing the test scores.

Limitations of each model. Logistic regression is linear (cannot follow patterns like experience rising then falling) and its coefficients are hard to read with correlated features; forests and boosted trees overfit unless leaves/depth are limited and their probabilities are not perfectly calibrated; the network has no extra information to learn from hand-made features (about 18,700 rows after the early-stopping split) and depends on the seed (fixed).

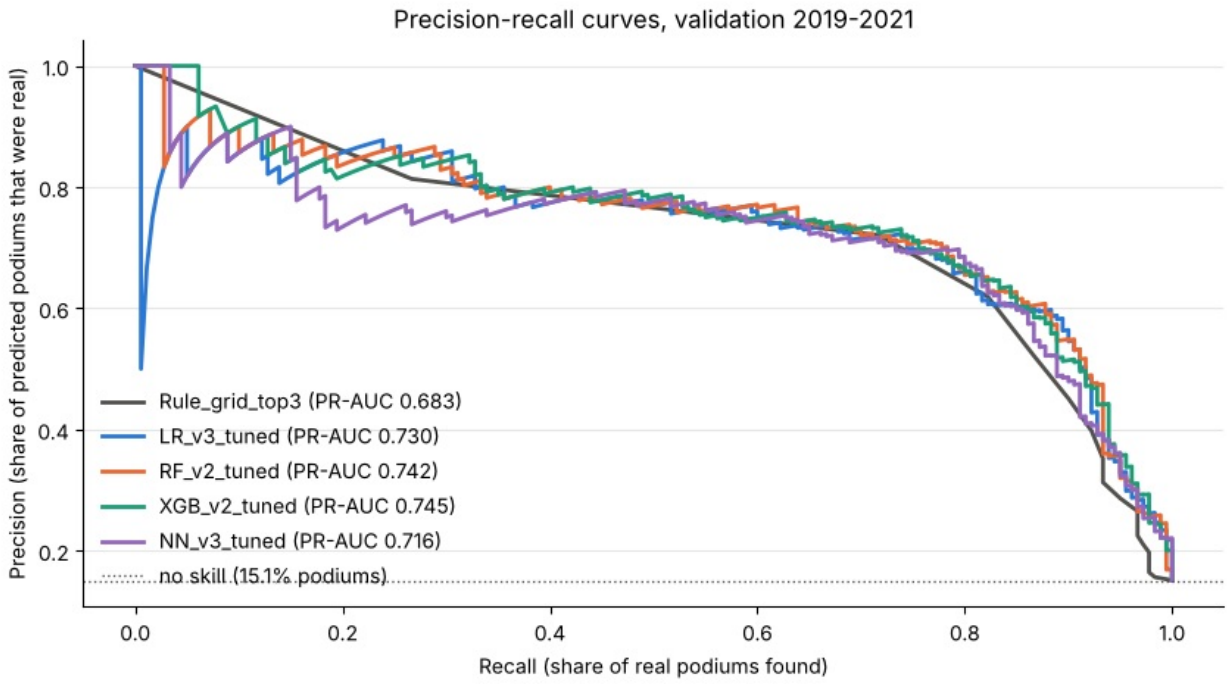


Figure 11. Precision-recall curves on validation (2019–2021), best attempt of each family; dotted line = no skill (15.1% podiums).

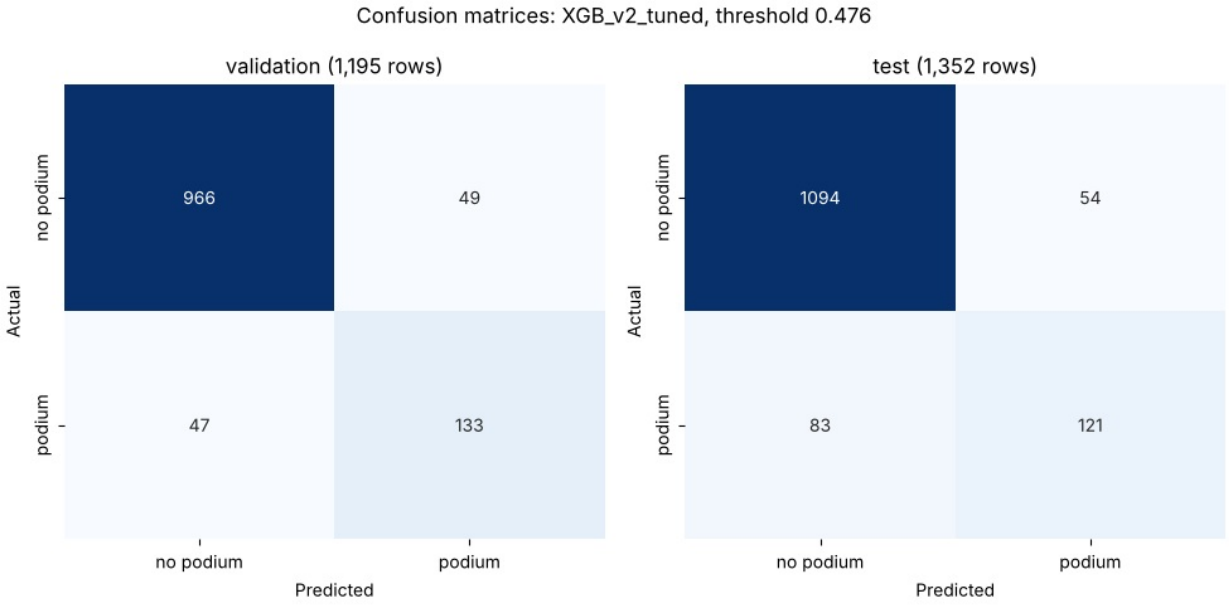


Figure 12. Confusion matrices of the final model at threshold 0.476, validation (left) and test (right).

ROC-AUC is above 0.9 for every model including the rule (notebook 10.10), because the back of the grid is easy to rule out; PR-AUC shows the harder part — picking the three podium drivers among the front runners — which is why it is the main metric.

7. Experiments

Experiments use the best scikit-learn model (`RF_v2_tuned`, validation PR-AUC 0.743) — plus the final XGBoost where the question concerns it — and are scored on **validation only**. A difference of 0.005 PR-AUC or less was fixed in advance as "the same"; random-forest results are averaged over 3 seeds where noted.

Feature-group ablation. Removing the **grid** group costs the most (0.743 → 0.719, −0.023): the grid after penalties is the information closest to the start and exists in every season. Driver form (−0.009), standings (−0.007), constructor form (−0.006) and qualifying (−0.004) cost little alone because they overlap (they all measure how strong car and driver are). Without the context group the model is slightly *better* (+0.004) — within noise for 180 validation podiums, consistent with Q2 (no home effect).

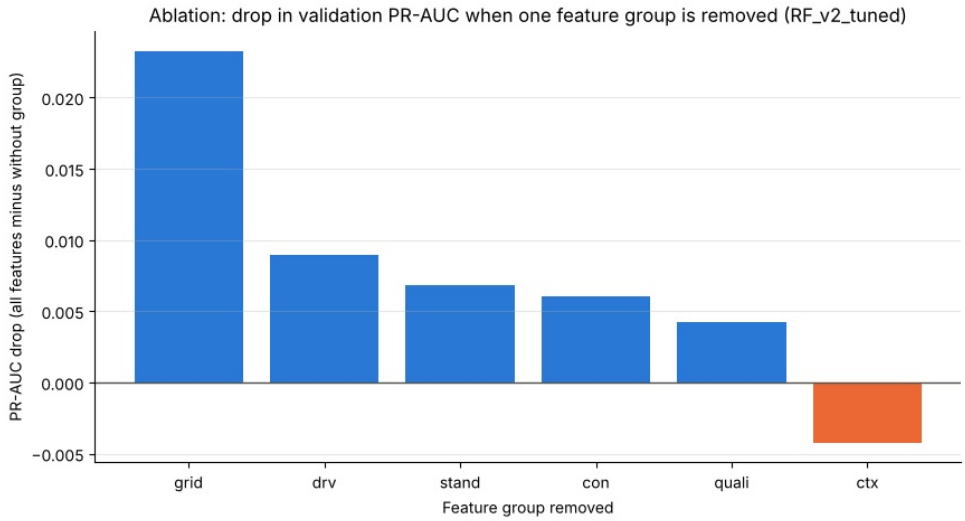


Figure 13. Ablation: drop in validation PR-AUC when one feature group is removed (orange = slightly better without it).

Experiment	Setting	Validation PR-AUC	What it shows
1. Imputer + scaler	fitted on train only (our Pipeline)	0.7292	clean estimate
	fitted on all seasons before the split	0.7294	9 of 20 medians change (e.g. team points 6 → 8): the future shapes the transformation; tiny here, larger after rule changes
2. Oversampling podiums to 50/50	none	0.743	reference
	after the split (train only)	0.731	no gain — changes probabilities more than ranking
	before the split	0.934	fake: "validation" becomes 53% podiums with copied rows
3. Training years (Decision 4)	all seasons 1950–2018 (64% of rows without qualifying)	XGB 0.744 / RF 0.738 / LR 0.729	main setting
	1994–2018	XGB 0.742 / RF 0.737 / LR 0.720	no gain
	2003–2018 (complete qualifying)	XGB 0.732 / RF 0.745 / LR 0.726	worse for the final model; slightly better for the forest
4. Missing values (Decision 22)	median + indicators vs mean, no indicators, −1, KNN, dropping rows with gaps	RF: all within 0.005 (0.735–0.740)	LR: dropping rows −0.008, KNN −0.007, no indicators −0.004; XGBoost native vs imputer: same (0.744 vs 0.746) → keep median + indicators, keep all rows
5. Withdrawn drivers (Decision 8)	drop (current) vs keep the 322 W training rows	0.738 vs 0.738	+0.0002: no change → keep removing them

Training years in detail (11.5). We built the same features for every season and trained from 1950, 1960, … 2010 (always up to 2018). The final model is best with all seasons and drops steadily when training starts later (0.732 from 2003, 0.708 from 2010); the forest peaks at 2003; logistic regression is flat. The forest's preference is not caused by tuning (re-tuned on 2003+, it scores the same). A bootstrap over the 60 validation races puts every difference inside chance (XGBoost: all seasons better in 84% of resamples, difference +0.010, interval −0.009 to +0.032; forest: 2003+ better in 92%, −0.006, interval −0.016 to +0.002). Conclusion: using the whole history, as the TAs asked, costs nothing reliable and helps the final model.

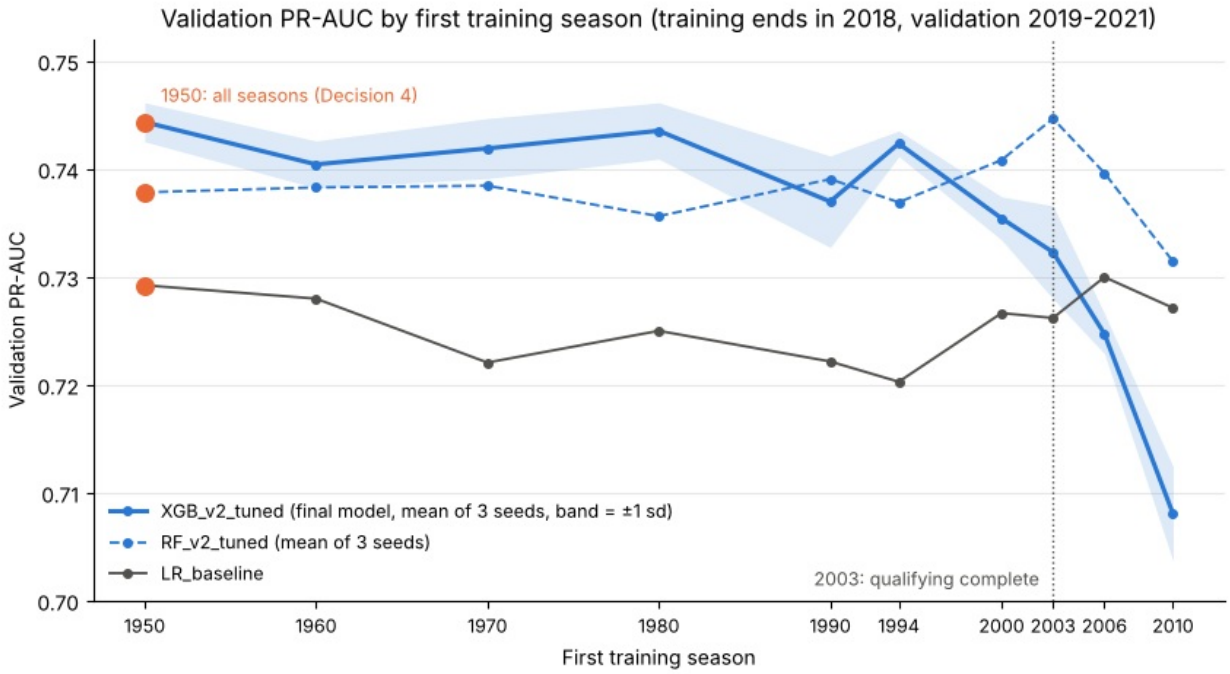


Figure 14. Validation PR-AUC by first training season (training always ends in 2018): the final model (solid) is highest with all seasons (orange dots), the forest (dashed) peaks at 2003, logistic regression (grey) is flat.

Withdrawn drivers in detail (11.7). A withdrawal can happen after qualifying (129 of the 327 W rows had a grid slot), so removing them needed evidence. Their features were rebuilt “as of the race” with the Section 8 rules (a check reproduces all 20 features of the 23,173 normal train + validation rows exactly). Keeping them changes validation PR-AUC by +0.0002 (+0.0010 without the 2005 US GP, when 14 cars withdrew) — the same. The model would still rate Leclerc (pole, Monaco 2021, did not start) at 0.58: such events cannot be predicted before the start.

8. Explainability

> **Important:** “important to the model” does **not** mean “causes podiums”. These methods show which inputs the model **relies on** to make its predictions. The features are correlated with each other and with things the data does not contain (car speed, budget, team quality); a good grid slot is a result of a fast car and driver, not a cause we could change.

Global — permutation importance (final model, validation): shuffling `grid_position` costs **0.146** PR-AUC, `quali_position` 0.045, team and driver podium rate in the last 10 races 0.017 and 0.016, the driver’s championship position 0.012, the front-row flag 0.009; pit-lane, home race and field size about 0.

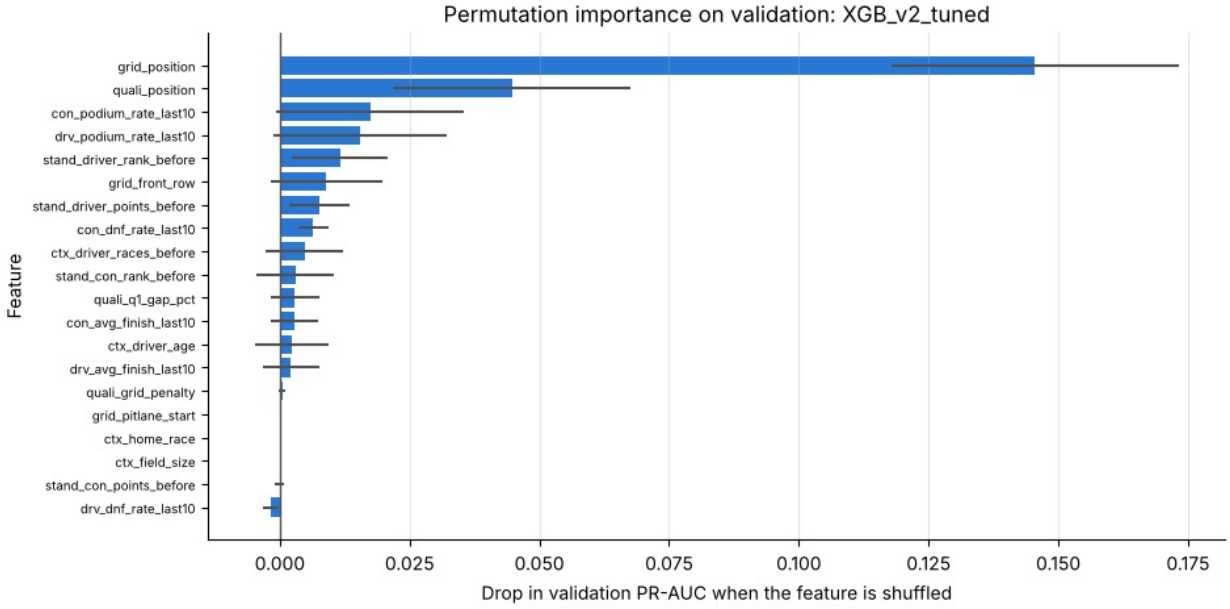


Figure 15. Permutation importance of the final model on validation (mean drop in PR-AUC over 10 shuffles, ± 1 standard deviation).

Global — SHAP (exact TreeSHAP of the final XGBoost model on 500 validation rows): the input that moves predictions most is the **grid position** (mean |SHAP| about 1.22 log-odds), then team podium rate (0.38), driver podium rate and qualifying position (0.24 each), then the championship ranks. A front slot (blue = low value) pushes towards a podium, a slot further back pushes away. The same groups lead as in the permutation importance; qualifying position ranks lower in SHAP because it is 0.97 correlated with grid position and SHAP shares the credit between them.

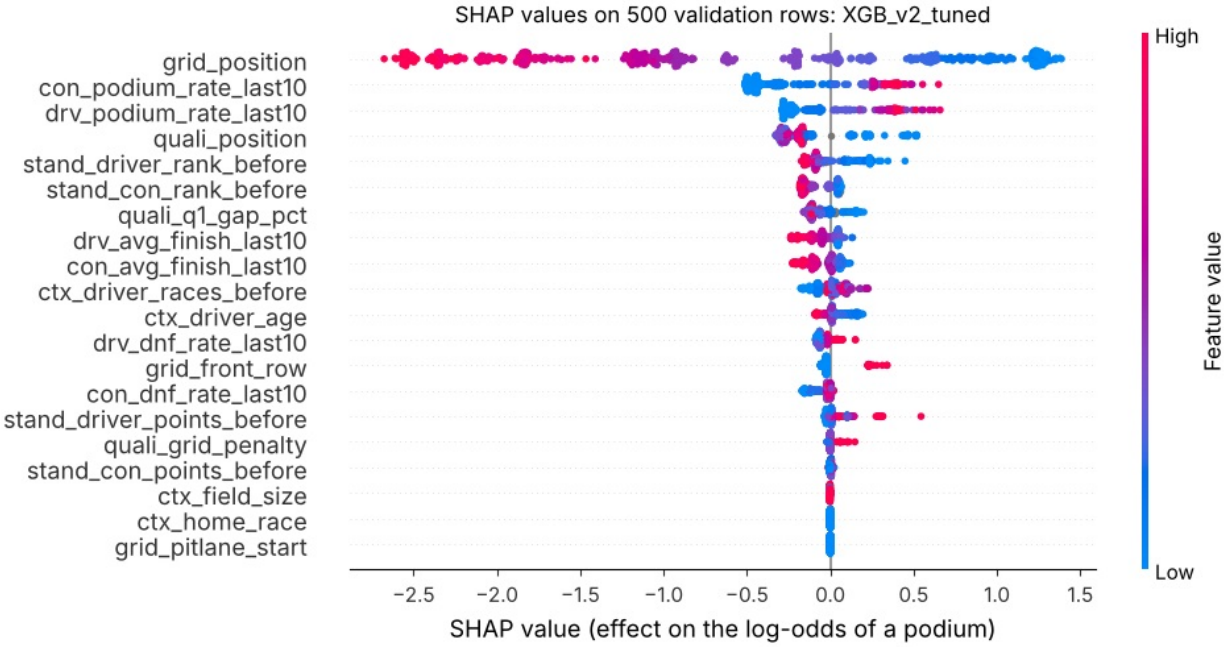


Figure 16. SHAP beeswarm (XGB_v2_tuned, 500 validation rows): one dot per row; right = towards a podium; colour = feature value (red high, blue low).

Local — LIME (final model) on two validation cases:

- **Surprise podium: Daniil Kvyat, 2019 German GP** — started 14th, model probability 0.010, finished 3rd in a chaotic wet race. LIME: grid 12–17 (−0.062), no team podium in the last 10 races (−0.031), not on the front row (−0.025), no driver podium (−0.024). The model gave him a low chance for the right pre-race reasons; the race events that produced

the podium are, by design, not features.

- **Missed favourite: Valtteri Bottas, 2020 Sakhir GP** — pole, probability 0.822, finished 8th after a pit-stop mix-up by his team. LIME: grid ≤ 6 (+0.152), team podium share above 0.23 (+0.044), driver share above 0.2 (+0.038), front row (+0.026).

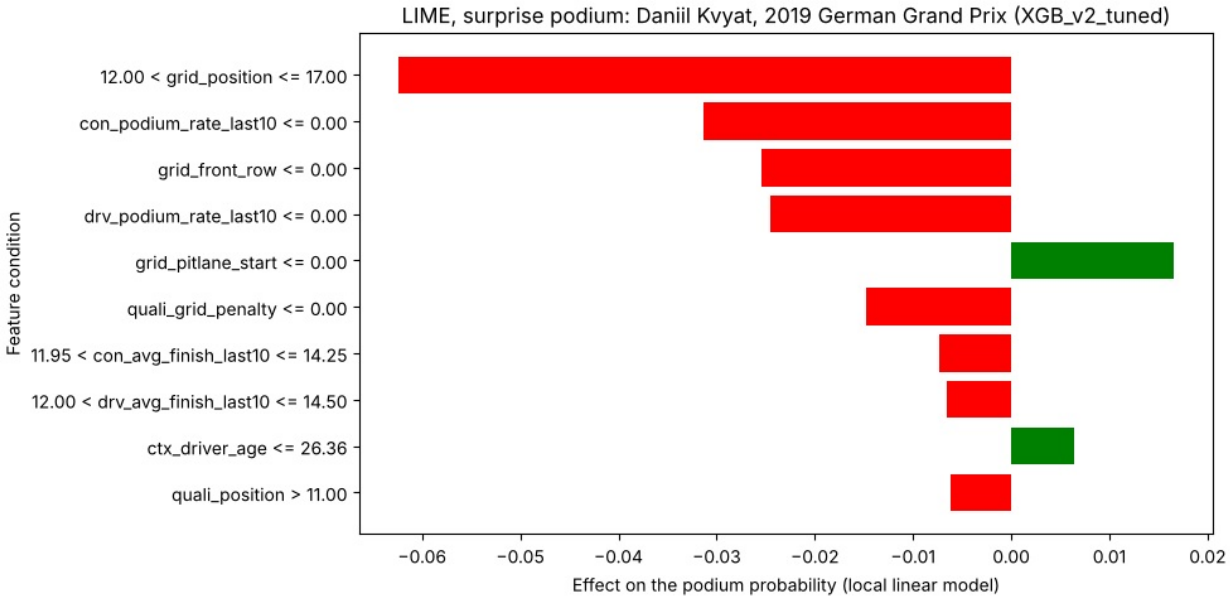


Figure 17. LIME explanation of the final model for the surprise podium (Kvyat, 2019 German GP): red pushes the probability down, green up.

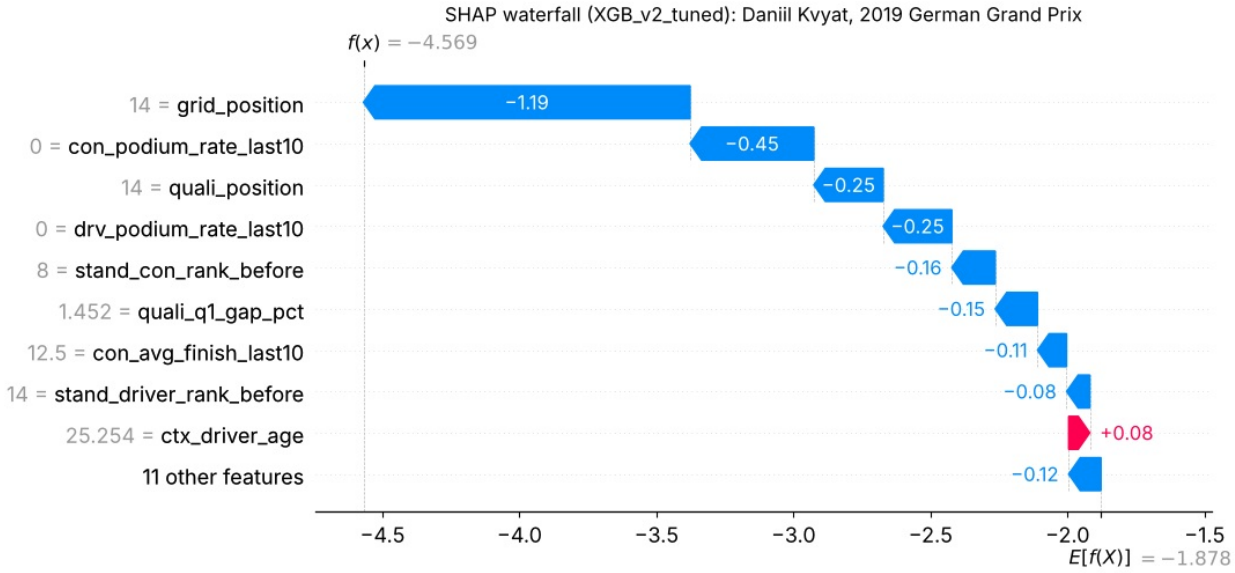


Figure 18. Local SHAP waterfall (final model) for the same driver: from the average log-odds −1.88 to −4.57; grid 14th (−1.19) and no team podium in the last 10 races (−0.45) push down most — the same story as LIME.

9. Inference function

`predict_podium(record)` takes one driver-race as a dictionary of feature values, builds a one-row table with exactly the 20 features (anything missing or `None` becomes NaN), and returns the probability from the saved final Pipeline (XGBoost handles the NaN values as learned in training) and a yes/no answer at the validation threshold 0.476. It refuses unknown or post-race columns (e.g. `positionOrder`).

Demo	Input	Output
Complete record	Max Verstappen, 2021 Abu Dhabi GP (pole, all 20 features)	probability **0.843** → podium; equals the Pipeline's own prediction (assert)
Partially missing record	rookie, grid 6, no qualifying time, no driver history (7 features missing)	probability **0.223** → no podium; missing features listed

10. Known limitations

- **Missing qualifying in old seasons:** the model trains on all seasons (TA requirement), so qualifying is missing in 64% of training rows; the gaps are filled with "was missing" flags and the experiments show no loss, but the model learns qualifying effects mainly from 1994+ rows.

- **Era sensitivity:** eras differ strongly (reliability, field size, rules); there is no era feature (Decision 20), and the test seasons are the 2022 rules era — scores drop from validation (0.745) to test (0.711). Modelling the eras explicitly is planned for the next milestone.
- **Small validation set:** 180 podiums — differences below about 0.01 PR-AUC between models, start years or ablation settings are within noise (bootstrap, Section 7).
- **Race events are invisible by design:** crashes, safety cars, penalties, pit-stop errors and weather (e.g. Kvyat 2019, Bottas 2020) cannot be predicted before the start.
- **Field size:** `ctx_field_size` counts starters, so races with withdrawals show a slightly smaller field than the one known before the start (largest case: 2005 US GP, 6 vs 20); the 11.7 experiment includes the corrected count with no change in score.
- **Judgement calls:** status → mechanical mapping (partly TA-fixed, robust in a broad check), nationality dictionary, team-lineage map (Q3 only), duplicate rule, Indy 500 exclusion, pit-lane encoding, form window.
- **Sprint weekends 2021–22:** the qualifying table describes the sprint grid, not the Sunday grid.
- **Data quirks:** grid and qualifying almost never differ before 2004 (penalties probably not recorded); fastest-lap columns exist only from 2004; no engine-supplier or tyre data.
- **Small samples in Q1–Q3:** 10–26 front-row starts per circuit after 2014; about 135 starts per team in 2022–2024.

Appendix: decision log

#	Decision	Evidence
3	PR-AUC main metric, ROC-AUC secondary; F1, precision, recall reported	13.6% podiums (cell 4.9)
4	Model trains on all seasons (1950–2018); `QUALI_COMPLETE_YEAR = 2003` only marks the qualifying era	TA requirement; 11.5: final model 0.744 all seasons vs 0.732 from 2003, all gaps within chance (bootstrap)
5	Drop all rows of repeated driver-race pairs	grid can’t separate cars; first row follows finishing order
6	Exclude the Indianapolis 500	0.6% driver overlap, 99.8% American
7	Link renamed teams, Q3 only	6 vs 10 comparable teams; renames within noise
8	Drop non-starters (F/W), keep 87 pit-lane starters	entries vs starters (4.4), grid = 0 (4.6, 4.7); F known before the race (TA); keeping W: +0.0002 (11.7)
9	Sprints: use partly (Sunday grid, no sprint features)	grid = qualifying in 12–18% of 2021–22 sprint rows
10	Split 1950–2018 / 2019–2021 / 2022–2024	random split leaks (9.1); option table (9.2); 2020 check (9.3)
11	Status → mechanical mapping	TA ruling + component rule; strict vs broad Spearman 0.91/0.95
12	Nationality → country dictionary	43 nationalities asserted
13–14	Read CSVs as text; convert types in place; save clean.pkl	160,144 markers → NaN; conversions checked
15–16	Q1 minimum 10 starts; Q2 grid buckets + race bootstrap	circuits kept per minimum; bucket rates
17	Pit lane = last slot + 1 + flag	0 podiums from 87 pit-lane starts; confirmed by a TA
18	Form over last 10 races, restart after 1-year break, NaN if no history	window trial (8.5)
19	Constructor form by constructorId	0.3% of rows differ, same AUC (8.7)
20	No year/era feature	base rate follows field size; unseen test years
21	Final model XGB_v2_tuned, threshold 0.476	best validation PR-AUC (10.8)
22	Missing values: training median + “was missing” indicators in the Pipeline; keep all rows; XGBoost native	11.6: 6 options × 2 models; dropping rows / KNN worse for LR; all within 0.005 for RF